

EXPERIMENT 8

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Experiment 8 -- To study and implement Views

Aim : To study and learn the about the Views, grant or revoke submission

Question :8 Part A (i): Complex View

```
CREATE TABLE customer_master  
(  
    customer_id VARCHAR(5) PRIMARY  
KEY,    full_name VARCHAR(50) NOT  
NULL,    city VARCHAR(30)  
);
```

```
CREATE TABLE product_catalog  
(  
    product_id VARCHAR(5) PRIMARY  
KEY,    product_name VARCHAR(50)  
NOT NULL,    unit_price  
NUMERIC(10,2) NOT NULL  
);
```

EXPERIMENT 8

CREATE TABLE sales_orders

```
(  
    order_id SERIAL PRIMARY KEY,  
    product_id VARCHAR(5)  
        REFERENCES product_catalog(product_id),  
    quantity INT NOT NULL,  
    customer_id VARCHAR(5)  
        REFERENCES customer_master(customer_id),  
    discount_percent NUMERIC(5,2),  
    order_date DATE NOT NULL  
);
```

INSERT INTO customer_master

(customer_id, full_name, city)

VALUES

```
('C1', 'Amit Sharma', 'Delhi'),  
( 'C2', 'Priya Verma', 'Mumbai'),  
( 'C3', 'Ravi Kumar', 'Bangalore'),  
( 'C4', 'Neha Singh', 'Kolkata'),  
( 'C5', 'Arjun Mehta', 'Hyderabad');
```

INSERT INTO product_catalog

(product_id, product_name, unit_price)

VALUES

```
('P1', 'Smartphone X100', 25000),  
( 'P2', 'Laptop Pro 15', 65000),  
( 'P3', 'Wireless Earbuds', 5000),
```

EXPERIMENT 8

('P4', 'Smartwatch Fit', 30000),
('P5', 'Gaming Console', 45000);

```
INSERT INTO sales_orders
(product_id, quantity, customer_id, discount_percent, order_date)
VALUES
('P1', 2, 'C1', 5, '2025-09-01'),
('P2', 1, 'C2', 10, '2025-09-02'),
('P3', 3, 'C3', 0, '2025-09-03'),
('P4', 1, 'C4', 8, '2025-09-04'),
('P5', 1, 'C5', 12, '2025-09-05'),
('P1', 1, 'C2', 5, '2025-09-06'),
('P3', 2, 'C1', 0, '2025-09-07');
```

```
CREATE OR REPLACE VIEW vw_sales_report as
select t.order_id, cc.full_name, t.quantity, t.final_am as final_amount
FROM (
    select so.discount_percent, so.product_id, so.customer_id, so.order_id, so.quantity, (
        (pc.unit_price * so.quantity * so.discount_percent )/100
    ) as final_am
FROM sales_orders as so
JOIN product_catalog as pc
ON so.product_id = pc.product_id
) t
JOIN customer_master as cc
```

EXPERIMENT 8

ON cc.customer_id =t.customer_id

SELECT * from vw_sales_report

CREATE USER Exp_8_user with PASSWORD '1234';

GRANT SELECT ON vw_sales_report TO Exp_8_user;

**Question: Part A (ii) Materialized view for Category_report_generation SQL
Query:**

DROP TABLE IF EXISTS category_master CASCADE;

DROP TABLE IF EXISTS product_master CASCADE;

DROP TABLE IF EXISTS customer_master CASCADE;

DROP TABLE IF EXISTS order_details CASCADE;

CREATE TABLE category_master (

category_id VARCHAR(5) PRIMARY

KEY, category_name VARCHAR(50)

NOT NULL

);

CREATE TABLE product_master (

product_id VARCHAR(5) PRIMARY

KEY, product_name VARCHAR(50)

NOT NULL, category_id

VARCHAR(5)

REFERENCES category_master(category_id)

);

EXPERIMENT 8

```
CREATE TABLE customer_master (
```

```
    customer_id VARCHAR(5) PRIMARY KEY,  
    customer_name VARCHAR(50) NOT NULL  
);
```

```
CREATE TABLE order_details (
```

```
    order_id VARCHAR(5) NOT  
    NULL,    customer_id  
    VARCHAR(5)  
    REFERENCES customer_master(customer_id),  
    product_id VARCHAR(5)  
    REFERENCES  
    product_master(product_id),    quantity INT  
    NOT NULL,    unit_price NUMERIC(10,2)  
    NOT NULL  
);
```

```
INSERT INTO category_master
```

```
(category_id, category_name)
```

```
VALUES
```

```
('C1', 'Electronics'),
```

```
('C2', 'Clothing'),
```

```
('C3', 'Books');
```

EXPERIMENT 8

INSERT INTO product_master

(product_id, product_name, category_id)

VALUES

('P101', 'Laptop', 'C1'),

('P102', 'Headphones', 'C1'),

('P201', 'Jacket', 'C2'),

('P202', 'Shoes', 'C2'),

('P301', 'SQL Book', 'C3');

INSERT INTO customer_master

(customer_id, customer_name)

VALUES

('U1', 'Aman'),

('U2', 'Priya'),

('U3', 'Rahul'),

('U4', 'Simran');

INSERT INTO order_details

(order_id, customer_id, product_id, quantity, unit_price)

VALUES

('O101', 'U1', 'P101', 1, 5000),

('O102', 'U2', 'P102', 2, 1000),

('O103', 'U1', 'P201', 1, 3000),

('O104', 'U3', 'P202', 2, 2000),

('O105', 'U4', 'P301', 3, 500),

EXPERIMENT 8

```
('O106', 'U2', 'P101', 1, 5000),  
( 'O107', 'U3', 'P201', 1, 3000),  
( 'O108', 'U4', 'P102', 1, 1000);
```

```
DROP MATERIALIZED VIEW mw_category_report
```

```
EXPLAIN ANALYZE
```

```
CREATE MATERIALIZED VIEW mw_category_report
```

```
AS
```

```
select cm.category_id as  
CatID,  
  
cm.category_name as CatName,  
  
SUM(od.quantity) as TOTAL_PRODUCT,  
  
COUNT(DISTINCT od.order_id) as TOTAL_ORDERS,  
  
COUNT(DISTINCT od.customer_id) as Unique_Customer,  
  
SUM(od.quantity * od.unit_price) as TOTAL_REVENUE,  
  
ROUND( SUM(od.quantity * od.unit_price)/COUNT(DISTINCT od.order_id) ,2) as  
AVG_REVENUE,  
  
(CASE WHEN SUM(od.quantity * od.unit_price) >= 10000 THEN 'Excellent'  
  
WHEN SUM(od.quantity * od.unit_price) >= 6000 THEN 'Good'  
  
WHEN SUM(od.quantity * od.unit_price) >= 3000 THEN 'Average' ELSE 'Low'  
END) as Category_performance  
from Order_details as od  
JOIN product_master as pm  
ON od.product_id = pm.product_id  
JOIN category_master as cm
```

EXPERIMENT 8

ON cm.category_id = pm.category_id

GROUP BY cm.category_id, cm.category_name

SELECT * FROM mw_category_report

Learning Outcome

After completing this experiment, the learner will be able to:

1. **Understand and implement Views**
2. **Apply aggregate functions** such as COUNT() and DISTINCT to get unique customer;
3. **Use joins and filtering techniques** to combine data from multiple tables and generate meaningful reports.